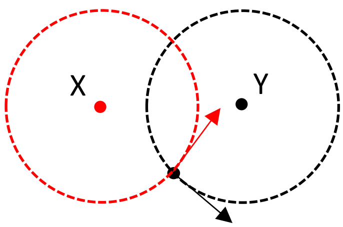
Qns	Answer	Marks
4(a)	Force per unit current unit length of wire Carrying a wire that is normal to the magnetic field.	**1**
(b)(i)1.	direction (dots above current, crosses below current) field strength stronger when nearer wire (higher density of dots/crosses nearer the current)	**1** **1**
(b)(i)2.	positive	**1**
(b)(ii)	Force is always perpendicular to the velocity of the particle <u>No displacement in the direction of force</u>, so <u>work done</u> by the force is <u>zero</u>, no change in kinetic energy so no change in speed.	**1** **1**
(b)(iii)	Done in vacuum, so no loss of kinetic energy due to no collision with air particles, so no change in speed. OR Weight of the particle is negligible compared to the magnetic force.	**1**
(c)	Observe that 3 cm, 4 cm and 5 cm form a right-angle triangle. So the magnetic flux density at point Z due to currents in X and Y are perpendicular to one another.  The magnetic flux density at Z is the <u>vector sum</u> of <u>magnetic flux densities</u> due to X and Y. $B = \frac{\mu_0 I}{2\pi d}$ $B_{resultant} = \sqrt{B_X^2 + B_Y^2}$ $= \sqrt{\left[\frac{(4\pi \times 10^{-7})(7.0)}{2\pi(3.0 \times 10^{-2})}\right]^2 + \left[\frac{(4\pi \times 10^{-7})(8.0)}{2\pi(4.0 \times 10^{-2})}\right]^2}$ $= 6.15 \times 10^{-5}$ T	 **1** **1**

Qns	Answer	Marks
5(a)(i)	At 22.5 ^{o}C, R_T = 1.6 kΩ (read from graph)	1